



Faculty of Electrical Engineering

Electronics lab

Signal processing using Audio

made by

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1 Objective

- Use the Bluetooth audio board on a FestoBoard to explore the real-time effects of analog filters on audio signals
- Learn how different filter types shape sound and evaluate the usability of the system as a learning tool.

2 User Guide

The board used in this lab is meant to feed an audio signal into analog circuits. It is designed to be placed on FestoBoards and features four pins. Below, a photo of the board, along with highlighted important features:

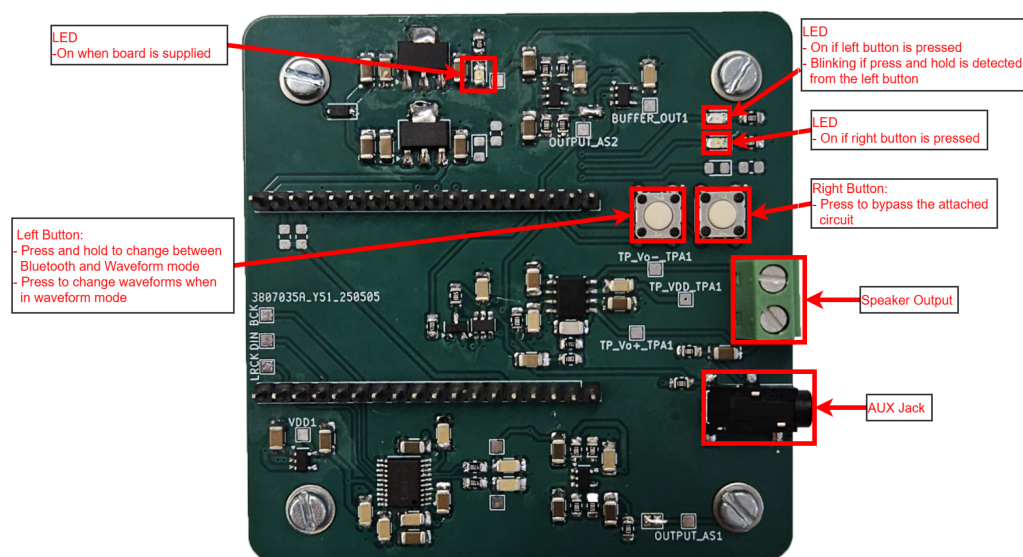
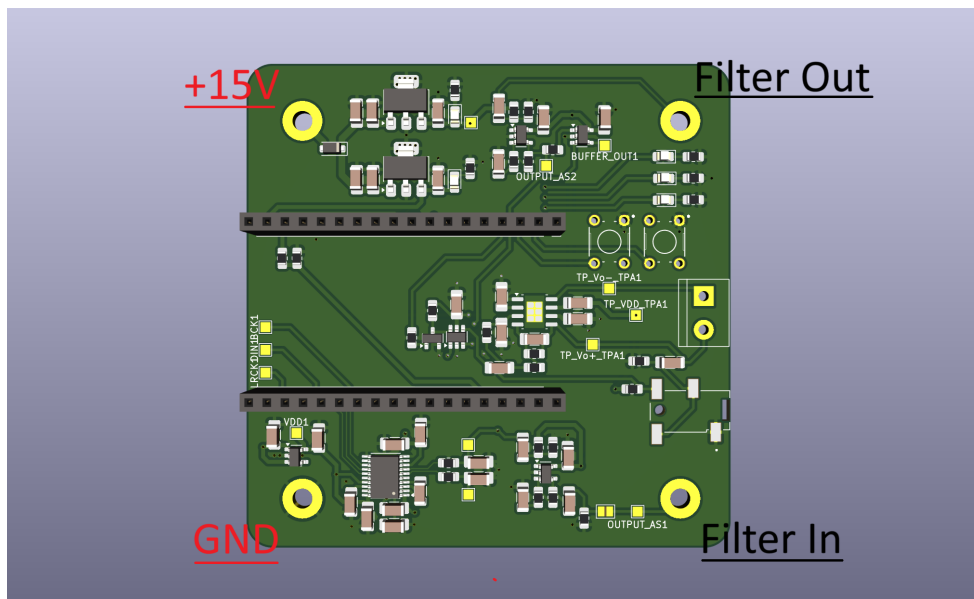


Abbildung 1: PCB Functionality

3 Setup Instructions

1. Configure the power supply: +15V with a 350mA current limit
2. Connect the audio board:



3. Power on the FestoBoard. If the PCB is connected properly, the topmost LED will turn on, as shown in the user guide section.
4. Connect your phone to the board that corresponds to your desk number
5. Test by playing some audio (e.g., sine sweep, music, voice)
Tip: Press and hold the left button to cycle between Bluetooth/Waveform mode. If in waveform mode, briefly pressing the same button will cycle through various waveforms. Pressing the right button will bypass the filter circuit.

4 Task No. 1: Filtering high-end frequencies

This task will focus on the effects of low-pass filters on audio signals.

1. Assemble the following active, second-order, low-pass filter:

$$R_1 = 3.2k, R_2 = R_3 = 10k, C_1 = 470nF, C_2 = 10nF$$

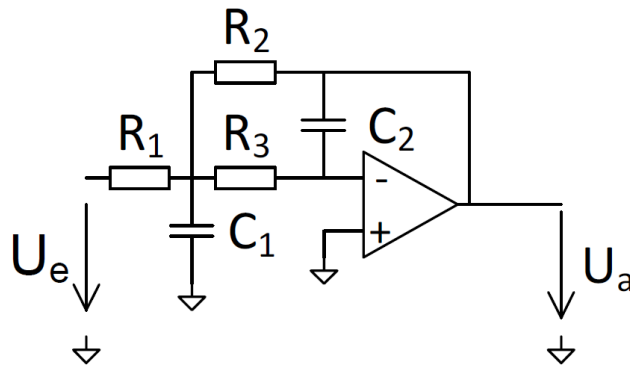


Abbildung 2: 2nd Order LPF, $f_c=230\text{Hz}$

2. Connect U_e to *Filter In* and U_a to the *Filter Out* pins of the board
3. Power on the FestoBoard, connect your phone, and verify connectivity
4. Play an audio sample (Ex. Bees Gees - Stayin' Alive, white noise, a baby crying).
5. Listen and analyse, what do you hear? Why is this happening? Argue your reasoning
Tip: Use the Bypass function to identify the effect
6. For a more controlled test, use the on-board waveform generator; try cycling between the waveforms

5 Task No. 2: Exploring Filter Orders and Roll-off

This task will explore various filter topologies and their effects.

There are different ways to implement filters. For this case study, three different types of low-pass filters will be used to study their roll-off regions.

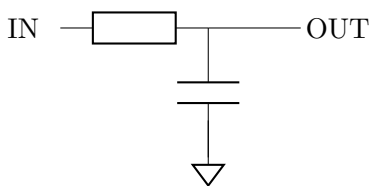


Abbildung 3: 1st-Order, Passive

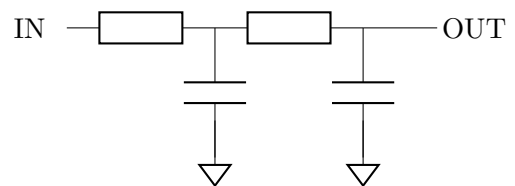


Abbildung 4: 2nd-Order, Passive

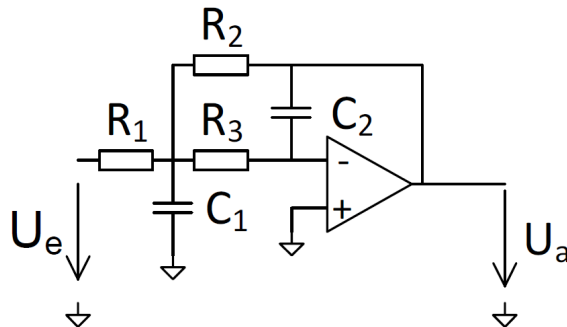


Abbildung 5: 2nd-Order, Active, same as Task 1

Dimension the circuits with a 230 Hz corner frequency. Use the 5th waveform type (press the right button 4 times). Listen to the output signal and notice the differences in roll-off. Which topology would be better to use? Why?

Tip: try building a 3rd-order LPF by connecting a 1st-order filter with a 2nd-order in series. What should be the order in which you connect them?

6 Task No. 3: Filtering an unwanted frequency component

This task will show how filters are used to remove unwanted information from a signal.

Say you have a sensor that sends you analog data at a rate of 1.5 kHz. Unfortunately, due to incompetent wiring from your Intern, you receive an additional parasitic, low-frequency component in your signal. This will heavily affect the precision of your measurement and ruin your career. Your task is to analyse the incoming signal and filter the unwanted component:

1. Power on the board and activate Waveform mode
2. Cycle to Waveform 4 (press the button 3 times)
3. Listen to this signal, what do you hear?
4. Analyse in detail: use the Math – > FFT function of the Oscilloscope to analyse the frequency spectrum
Tip: set the center as 10kHz and the maximum as 20kHz when setting the FFT
5. Conceptualise a second order filter to address the problem
Tip: frequency lies somewhere under your main signal in the spectrum, what does that mean? What would be the cutoff frequency of the filter?
6. Check your result by comparing the un-/bypassed signal path

6.1 Feedback Box

As this was a Focus Group, you are now asked to provide feedback on the exercises, setup, and the board itself.

Feedback:

What did you enjoy in this lab?

What could be improved?

Any additional comments?